Vibration Monitoring and Alignment Standard PS-VIB-01

Plant standard for all rotating machines

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Plant engineering

Read this manual before installing, operating or servicing the equipment. Keep it at the machine.

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1 Purpose

This plant standard sets how vibration is measured, which limits apply and what to do when they are exceeded. It applies to all rotating machines in Plant A and Plant B.

2 Fixed alarm limit

The plant alarm is raised when the vibration RMS value stays above 4.5 mm/s for 30 minutes. This is the boundary of zone C/D in ISO 10816-3 for medium-size machines on rigid foundations.

Zone	RMS (mm/s)	Action
A/B	< 2.8	Normal operation
C	2.8–4.5	Plan action at the next stop
D	> 4.5 for 30 min	Alarm: stop as soon as possible

3 Limits relative to the machine's own baseline

A fixed limit fits medium-size machines. Small machines (conveyors, test benches, robots) rarely reach 4.5 mm/s even when badly damaged, and rough machines (presses) can reach it early. Every machine therefore also has its own baseline, recorded after installation and after each repair.

Change vs baseline	Meaning	Action
+25%	Attention	Check trend weekly
+50%	Warning	Find the cause, plan action
+100%	Alarm	Repair at the next stop

4 Reading the signals

- Kurtosis above 4: impacts, typical of rolling-element bearing damage.
- Rising RMS with normal kurtosis: misalignment, looseness or unbalance.
- Rising temperature and power with normal vibration: cooling or lubrication problem.
- Always compare at the same speed and load.

5 Shaft alignment

Align couplings and pulleys with a laser tool after every motor or pump change and at the semi-annual service. Record the final values in the work order.

Speed	Offset	Angle
up to 1,500 rpm	0.08 mm	0.07 mm/100 mm
1,500–3,000 rpm	0.05 mm	0.05 mm/100 mm
above 3,000 rpm	0.03 mm	0.04 mm/100 mm

6 Predictive monitoring

PlantGuard scores every machine every minute from these signals and reports the risk of a breakdown within 72 hours. A risk of 70% or more counts as an alert. Alerts are checked by a person before any repair or order is made.